STUDENT MANAGEMENT SYSTEM

Java Lab 1 Report
Year 2 CSE

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1. Project Overview

Project Name: Student Management System
Language: Java (Swing GUI)
Database: MySQL via JDBC
IDE: NetBeans

Description:
The Student Management System is a Java based application designed to manage student records efficiently. It allows users to perform essential operations like adding, updating, deleting, and searching student records.

The system demonstrates **object-oriented programming**, **database integration**, and **user interface design** while solving real-world problems in educational management.

Purpose:

- Simplify student record management.
- Reduce errors from manual record-keeping.
- Provide a user-friendly interface for staff.
- Serve as a practical learning project applying Java programming concepts.

Impact:

- Increases efficiency in handling student data.
- Improves accuracy and reliability of records.
- Can be adapted for use in schools or institutions.

2. Participants & Roles

Name	Role & Responsibilities
224005559	Designed login GUI, input validation, database connectivity, and CRUD functionality using JDBC.
224004620	Designed main page GUI, managed layout, color theme selection, and assisted in ideation of the overall system.

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3. Core Functionality

3.1 User Authentication

- Login interface with username and password.
- Reset option for password or login details.
- Feedback messages for invalid login attempts.

3.2 Student Management

- Add, Update, Delete, and Search student records.
- Display all records in a table format.

3.3 Database Integration

- Stores data securely in MySQL
- Uses JDBC with Prepared Statement to prevent SQL injection.

3.4 Input Validation

- Ensures fields are correctly filled before submission.
- Disables buttons until valid input is provided.

3.5 String Processing

- Converts names to proper title case.
- Supports case-insensitive search and string operations.

4. System Workflow

1. User logs in using valid credentials.
2. On the main page, the user can select operations: Add, Update, Delete, Search for admin and same for teacher except that teacher can't delete a student.
3. Data input is validated for completeness and correctness.
4. Records are stored and updated in the database.
5. Users receive feedback messages for each operation.
6. System maintains data integrity and allows fast access to records.

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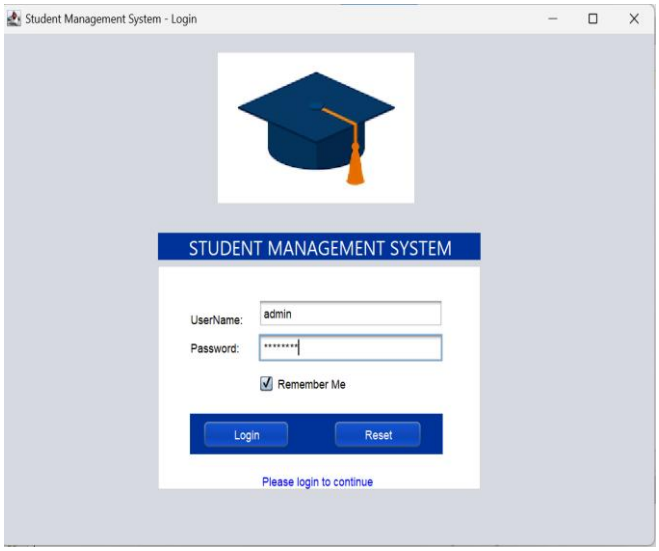
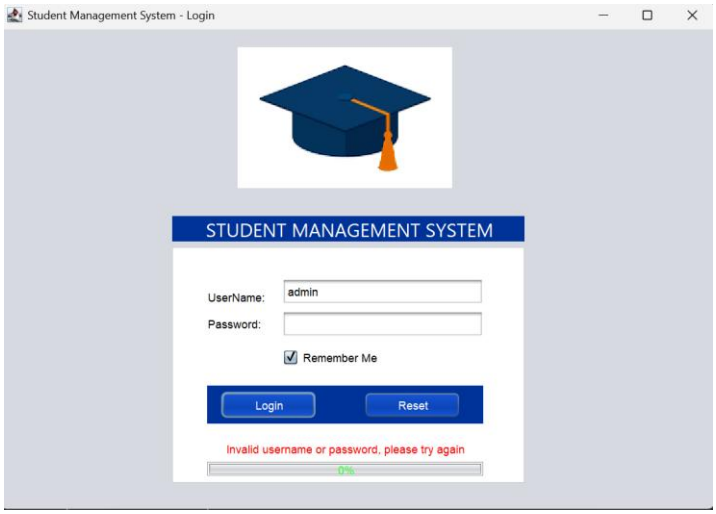
5. Screenshots

Login Page

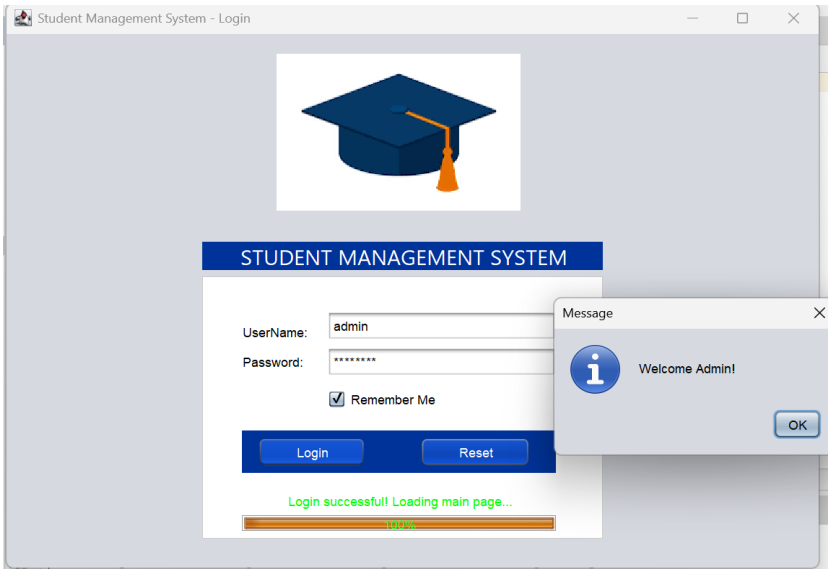


(while entering the credentials)

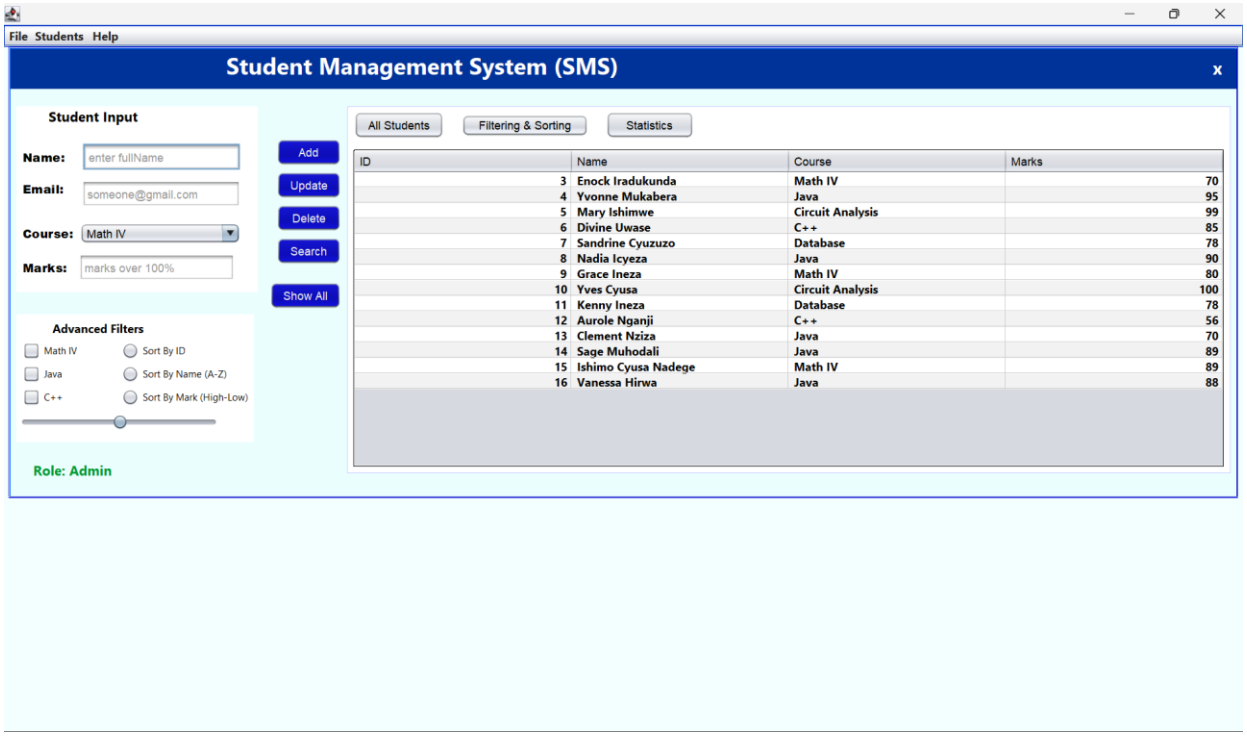


(feedback for wrong credentials)

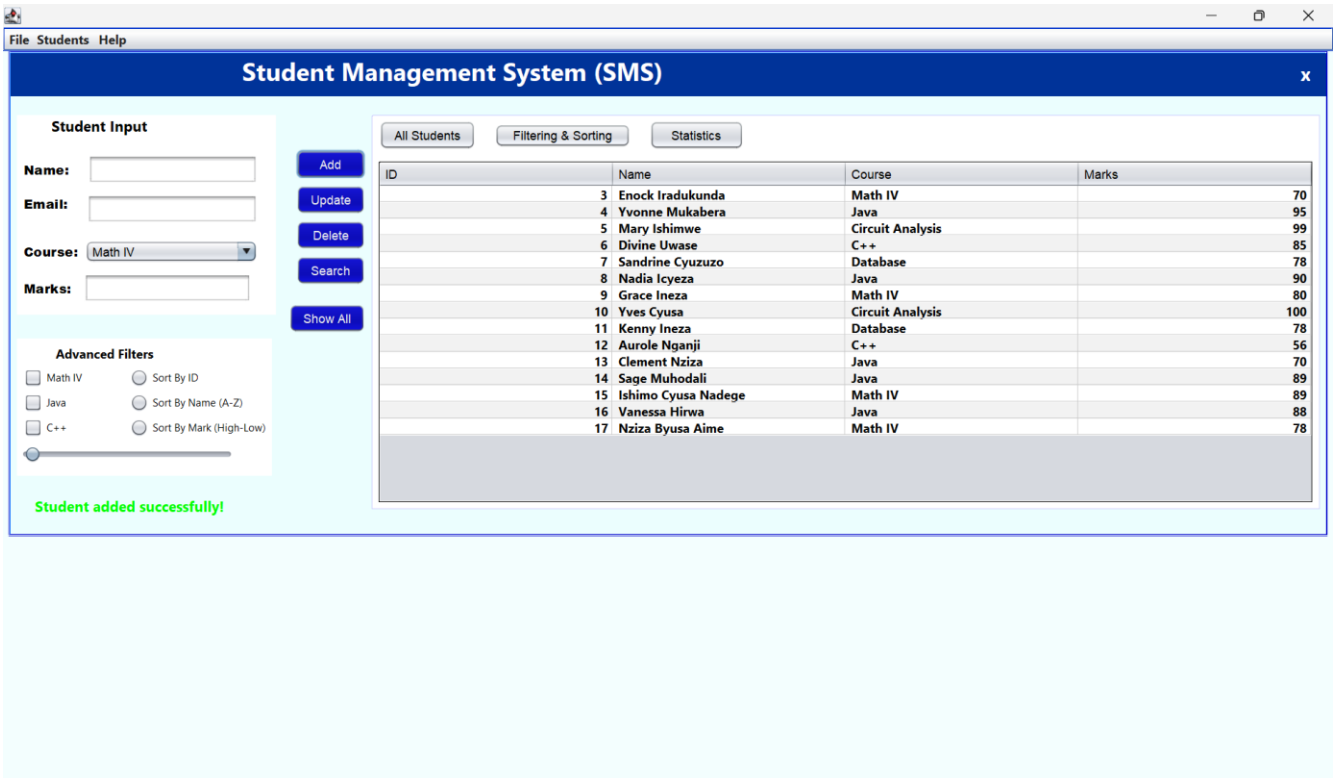
(feedback for correct credentials)



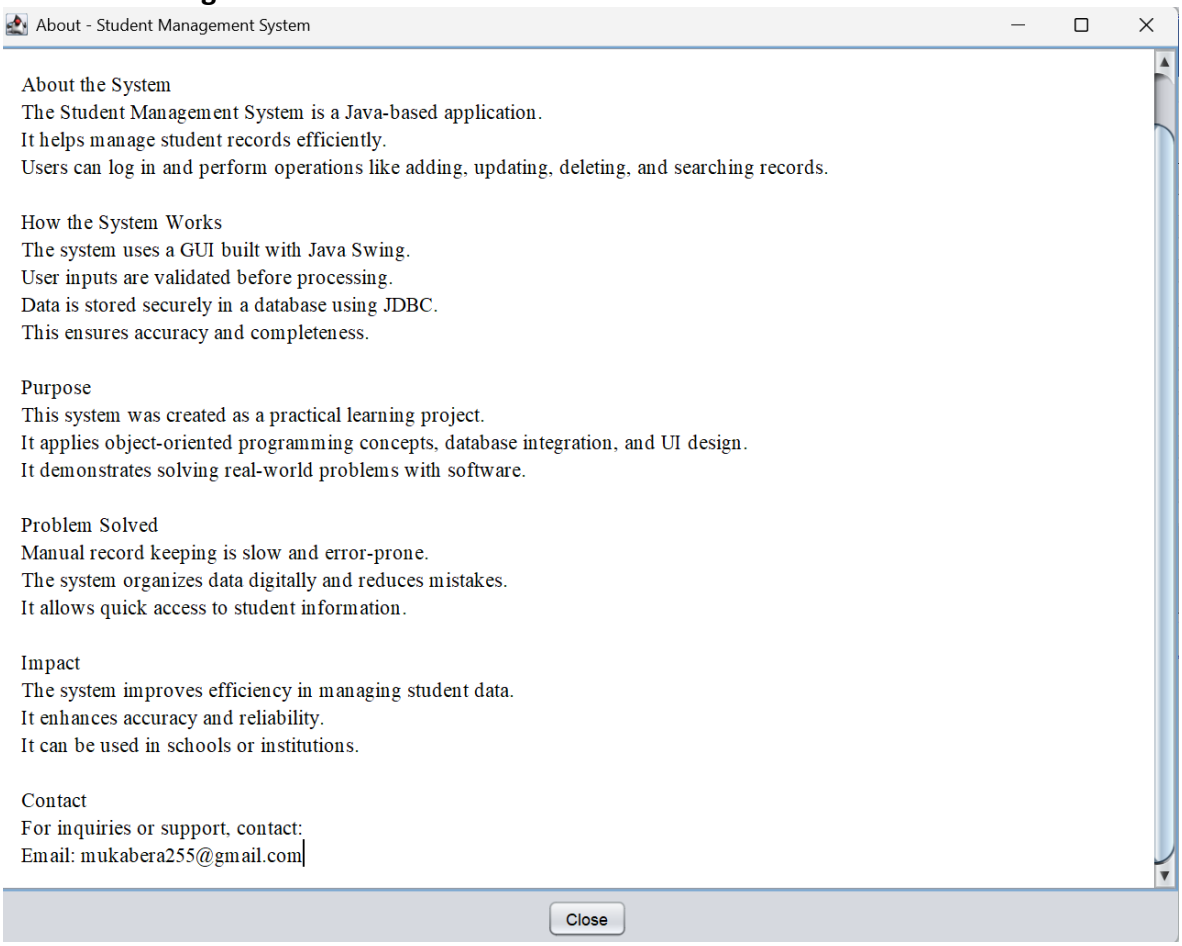
Main Page



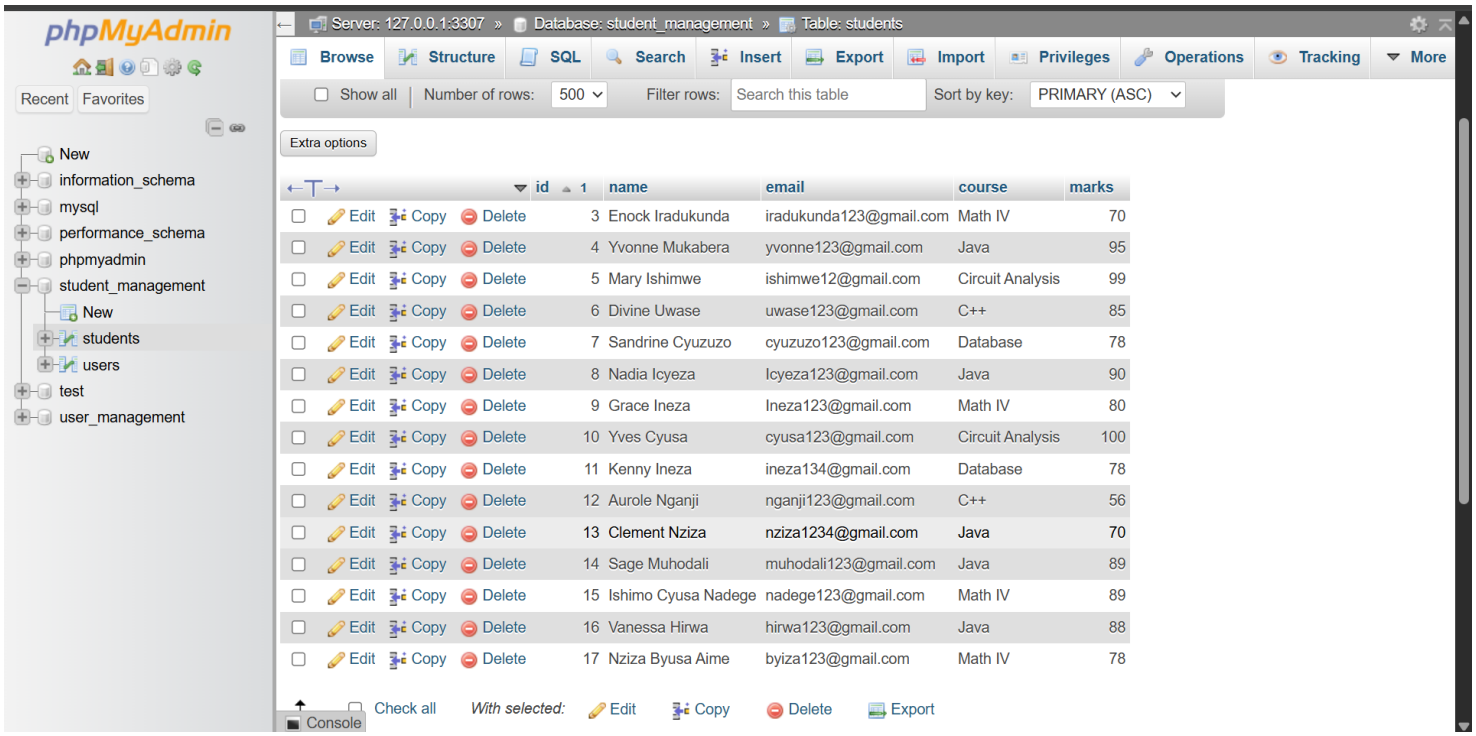
Add Student



About Page



Database



6. Challenges Faced

- Handling input validation efficiently.
- Integrating GUI components with the database.
- Ensuring data consistency during concurrent operations.